Data Analysis

BY THE END:

You can explain how data analysis uses data to answer questions and help people decide what to do.

Overview

A shop owner notices that one product sells out every week while another sits on the shelf. She wants to understand why and decide what to order next. This is where **data analysis** becomes useful.

What is data analysis?

Data analysis means looking at data, cleaning it, organizing it, and using it to find useful answers.

Data analysis may involve:

- Looking at how sales change
- Comparing results in different places
- Finding out why customers leave
- Preparing reports for managers

Data analysis often answers questions like:

Which products sold the most?

Why did the money earned go down in March?

Which customer group makes the most profit?

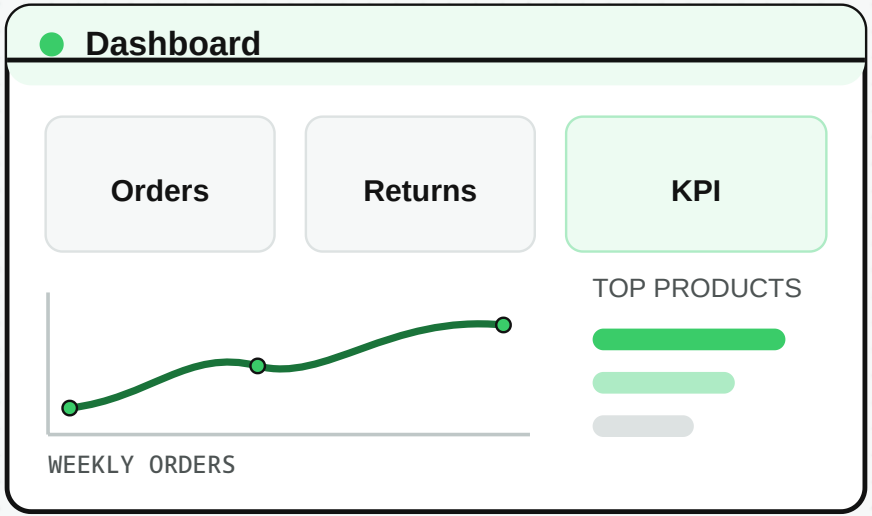
What data analysis can produce

The result depends on the question. Data analysis can produce:

- A finding that explains what happened
- A report that shows what the data found
- A metric that tracks something important
- A dashboard that keeps results visible over time

A KPI (key performance indicator) is a metric used to measure progress toward a goal.

Regular dashboards and reports are often called **business intelligence (BI).** Companies may use different names for this work, but the goal is the same: help people see useful information and make decisions.



Example

A wholesaler supplies shops in Mogadishu, Hargeisa, and Kismayo, and wants to know how each is doing.

- The analysis compares six months of orders and returns
- It finds that the Kismayo shops order the same amount as the others but send back much more unsold stock
- The result is shared in a report and a dashboard showing each city's weekly orders, returns, and top products

The analysis answered the manager's question and helped her decide what to order next.

